

Chapter 2: Measures of Location and Spread

A Level Statistics Notes · Rexponential

Why this chapter matters

Once data has been properly collected, the next step is to summarise it with a small number of representative values. This chapter covers the two core ideas — where the data is centred, and how spread out it is — for both raw and grouped data, plus the linear coding technique that simplifies awkward calculations by transforming the data first and reversing the transformation at the end. It also covers how to spot outliers and describe the skewness of a distribution using these same measures. These are the summary statistics used throughout the rest of the course, from comparing distributions to setting up hypothesis tests later on, so fluency here pays off in every chapter that follows.

2.1 Measures of central tendency

Concept

There are three common measures of central tendency — the mean, median and mode — each describing the “centre” of a data set in a different way.

The **mean** uses every value in the data set and is the most commonly used average, but it is sensitive to extreme values. The **median** is the middle value when the data is ordered, and is unaffected by extreme values, making it more useful for skewed data. The **mode** is the most frequently occurring value, and is the only average that can be used with qualitative data.

For **grouped data**, individual values are lost, so the mean is calculated using the **midpoint** of each class as an approximation for every value in that class — this only ever gives an *estimate* of the true mean, never an exact value. The median of grouped data is estimated using **linear interpolation**: assuming the values are evenly spread within the median class, and finding where the middle position of the data set, $n/2$, falls within it.

Key vocabulary

Term	Meaning
Mean	The sum of all values divided by how many there are
Median	The middle value of ordered data
Mode	The most frequently occurring value
Midpoint	The average of a class's lower and upper boundaries, used to represent every value in that class
Class boundary	The true upper or lower limit of a class, halfway between the end of one class and the start of the next

Procedure

1. For raw data: sum all the values and divide by n .
2. For data in a frequency table: multiply each value by its frequency, sum these, and divide by the total frequency.
3. For grouped data: replace each class with its midpoint, then apply step 2 using midpoints in place of exact values — state the answer as an *estimate of the mean*.
4. For a grouped median: identify the class containing position $n/2$ using the cumulative frequencies, then interpolate within that class.

Key formulas

For raw data:

$$\bar{x} = \frac{\sum x}{n}$$

For data in a frequency table:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

To estimate the median of grouped data:

$$\text{Median} = L + \left(\frac{\frac{n}{2} - F}{f_m} \right) \times h$$

where L is the lower class boundary of the median class, F is the cumulative frequency before the median class, f_m is the frequency of the median class, and h is the class width.

Worked example

EXAMPLE 2.1

The table shows the ages of 60 members of a gym.

Age (years)	10–19	20–29	30–39	40–49	50–59
Frequency	8	22	18	9	3

(a) Estimate the mean age. (b) Estimate the median age.

Step 1 (part a). Use class boundaries 9.5–19.5, 19.5–29.5, etc., giving midpoints 14.5, 24.5, 34.5, 44.5, 54.5. **M1**

$$\bar{x} = \frac{8(14.5) + 22(24.5) + 18(34.5) + 9(44.5) + 3(54.5)}{60} = \frac{1780}{60} = 29.7$$

A1

Step 2 (part b). Cumulative frequencies: 8, 30, 48, 57, 60. Here $n/2 = 60 \div 2 = 30$, and the cumulative frequency reaches exactly 30 at the end of the 20–29 class, so the median lies in that class. The median class is therefore 20–29 (boundaries 19.5 to 29.5), with $F = 8$, the cumulative frequency before it. **M1**

$$\text{Median} = 19.5 + \left(\frac{30 - 8}{22} \right) \times 10 = 19.5 + 10.0 = 29.5$$

A1

Answer (a) Estimated mean = 29.7 years. (b) Estimated median = 29.5 years.

EXAM TIP

Always write “estimate” when the mean or median comes from grouped data — examiners specifically look for this word (or equivalent phrasing) and it is often its own mark. Losing it is one of the most common ways to drop an otherwise fully correct answer.

2.2 Measures of spread

Concept

A measure of location alone can hide very different data sets — two classes could have the same mean mark but very different spreads of ability. The **range** (max – min) is the simplest measure but is easily distorted by a single extreme value. The **interquartile range (IQR)**, $Q3 - Q1$, describes the spread of the middle 50% of the data and is not affected by extreme values. **Variance** and **standard deviation** use every value in the data set, measuring the average squared (and then square-rooted) distance from the mean, and are the measures of spread used almost everywhere else in the course, including later in hypothesis testing.

Key vocabulary

Term	Meaning
Range	The difference between the largest and smallest values
Interquartile range (IQR)	$Q3 - Q1$: the range of the middle 50% of the data
Variance	The mean of the squared deviations from the mean
Standard deviation	The square root of the variance, in the same units as the original data

Procedure

1. Find Σx and Σx^2 (or Σfx and Σfx^2 for frequency data) — most calculators can calculate both directly using their statistics mode (sometimes labelled “SD” or “STAT” mode).
2. Find the mean using $\Sigma x \div n$ (or $\Sigma fx \div \Sigma f$).
3. Substitute into the computational formula for variance, using Σx^2 rather than working out each $(x - \text{mean})^2$ by hand.
4. Take the square root of the variance to find the standard deviation, and state the units.

Key formulas

For raw data:

$$\sigma^2 = \frac{\Sigma(x - \bar{x})^2}{n} = \frac{\Sigma x^2}{n} - \bar{x}^2$$

For data in a frequency table:

$$\sigma^2 = \frac{\Sigma fx^2}{\Sigma f} - \bar{x}^2$$

Then, to find the standard deviation:

$$\sigma = \sqrt{\sigma^2}$$

Worked example

EXAMPLE 2.2

A sample of 10 journey times (in minutes) gives $\Sigma x = 250$ and $\Sigma x^2 = 6390$. Find the standard deviation of the journey times.

Step 1. Find the mean: $250 \div 10 = 25$. **M1**

Step 2. Substitute into the variance formula.

$$\sigma^2 = \frac{6390}{10} - 25^2 = 639 - 625 = 14$$

M1 A1

Step 3. Take the square root to find the standard deviation.

$$\sigma = \sqrt{14} = 3.74 \text{ minutes (3 s.f.)}$$

A1

Answer Standard deviation ≈ 3.74 minutes.

EXAM TIP

Questions very often give Σx and Σx^2 directly (exactly as above) rather than a raw list of values, specifically to test whether you know the computational formula rather than only how to use a calculator's statistics mode. Make sure you can substitute into it fluently without a data set in front of them.

2.3 Coding data

Concept

When raw data values are large or awkward (e.g. house prices, or temperatures given to several decimal places), it is easier to transform every value using a simple linear rule. You then do the calculation on the transformed ("coded") data, and convert the result back to the original units at the end. This is called **coding**. Crucially, adding or subtracting a constant shifts the mean but leaves the spread unchanged, while multiplying or dividing scales both the mean and the spread — this asymmetry is the single most important thing to remember about coding.

Key vocabulary

Term	Meaning
Coding	Transforming data with a linear rule to simplify calculation
Linear transformation	A rule of the form $y = (x - a) / b$ (or equivalently $x = a + by$) applied to every value

Procedure

1. Choose constants a and b to make the transformed values simple to work with (often to remove a large shared offset and a common factor).
2. Apply the coding rule to every data value (or just use the given coded values, if the question provides them).
3. Calculate the mean and standard deviation of the *coded* data using the methods from 2.1 and 2.2.
4. Reverse the transformation: multiply back through by b and add a to recover the real mean; multiply the coded standard deviation by $|b|$ only (never adjust the standard deviation by a).

Key formulas

If every value is coded as $y = (x - a) / b$:

$$\bar{x} = a + b\bar{y}$$

$$\sigma_x = b\sigma_y$$

Worked example

EXAMPLE 2.3

A scientist records 8 temperature readings and codes them using $y = (x - 100) / 5$. The coded values have mean $\bar{y} = 3.4$ and standard deviation $\sigma_y = 1.2$. Find the mean and standard deviation of the original temperature readings.

Step 1. Reverse the coding for the mean: multiply by 5, then add 100. **M1**

$$\bar{x} = 100 + 5(3.4) = 117$$

A1

Step 2. Reverse the coding for the standard deviation: multiply by 5 only — do not add 100. **M1**

$$\sigma_x = 5 \times 1.2 = 6$$

A1

Answer Original mean = 117, original standard deviation = 6.

EXAM TIP

The most common error by far is adding the 100 back onto the standard deviation as well as the mean. Frame it physically: shifting every value by the same constant moves the whole data set but doesn't stretch or squash it, so the spread cannot change — only *scaling* (the $\div 5$ or $\times 5$ part) affects spread.

2.4 Outliers and skewness

Concept

An **outlier** is a value that is unusually far from the rest of the data. The most common rule (though a question may specify a different one) defines an outlier as any value more than $1.5 \times \text{IQR}$ beyond Q_1 or Q_3 . Outliers matter because they can distort the mean and standard deviation, which is part of why the median and IQR are preferred for skewed or outlier-heavy data.

Skewness describes the asymmetry of a distribution. In a symmetric distribution the mean, median and mode coincide. In a **positively skewed** distribution the tail stretches to the right (higher values), pulling the mean above the median, which is above the mode. In a **negatively skewed** distribution the tail stretches to the left, and the inequalities reverse.

Key vocabulary

Term	Meaning
Outlier	A value unusually far from the rest of the data (commonly beyond $Q_1 - 1.5 \times \text{IQR}$ or $Q_3 + 1.5 \times \text{IQR}$)
Positive skew	Mean $>$ median $>$ mode; tail stretches to the right
Negative skew	Mean $<$ median $<$ mode; tail stretches to the left
Symmetric distribution	Mean = median = mode

Procedure

1. Check the question for its own definition of an outlier — if none is given, use the $1.5 \times \text{IQR}$ rule above.
2. Calculate the lower and upper boundaries using the formulas below.
3. Any value below the lower boundary or above the upper boundary is an outlier.
4. To describe skewness, compare the mean and median (or look at a box plot's whisker lengths and where the median line sits within the box) and state the direction using the inequalities above.

Key formulas

$$\text{Lower boundary} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper boundary} = Q_3 + 1.5 \times \text{IQR}$$

Worked example

EXAMPLE 2.4

A data set has $Q1 = 12$, $Q3 = 30$, mean = 24.1 and median = 21. (a) A value of 55 is recorded. Use the $1.5 \times \text{IQR}$ rule to determine whether it is an outlier. (b) Comment on the skewness of the distribution.

Step 1 (part a). $\text{IQR} = 30 - 12 = 18$. **M1**

$$\text{Upper boundary} = 30 + 1.5 \times 18 = 57$$

A1

Since $55 < 57$, it is **not** an outlier. **A1**

Step 2 (part b). Mean (24.1) > median (21), so the distribution is positively skewed. **B1**

Answer (a) Not an outlier, since 55 is below the upper boundary of 57. (b) Positively skewed, since the mean exceeds the median.

EXAM TIP

Exam boards do not all use the same outlier rule — some define it using the mean $\pm k \times$ standard deviation instead of $1.5 \times \text{IQR}$. Check the question for a stated definition every time rather than defaulting automatically to $1.5 \times \text{IQR}$, since applying the wrong rule when one is given loses every mark that follows.

Chapter summary

Idea	Key point
Mean	$\Sigma x \div n$ (or $\Sigma fx \div \Sigma f$); for grouped data, an estimate using midpoints
Median (grouped)	Linear interpolation within the class containing position $n/2$
Variance / standard deviation	$\Sigma x^2/n - \text{mean}^2$, then square-root for standard deviation
Coding	Adding/subtracting shifts the mean only; multiplying/dividing scales both mean and spread
Outliers	Beyond $Q1 - 1.5 \times \text{IQR}$ or $Q3 + 1.5 \times \text{IQR}$, unless the question gives another rule
Skewness	Compare mean and median: mean > median $\rightarrow$ positive skew; mean < median $\rightarrow$ negative skew

Common exam mistakes

- Omitting the word “estimate” when quoting a mean or median calculated from grouped data.
- Adding the coding constant a back onto the standard deviation as well as the mean.
- Defaulting to the $1.5 \times \text{IQR}$ outlier rule when the question has specified a different one.
- Confusing which class is the median class when a cumulative frequency lands exactly on $n/2$.

You should now be able to

- Calculate the mean, median and mode from raw, frequency, and grouped data.
- Calculate variance and standard deviation using the computational formula, from raw values or from given Σx and Σx^2 .
- Code a data set with a linear transformation and correctly reverse the transformation for both the mean and the standard deviation.
- Identify outliers using the $1.5 \times \text{IQR}$ rule (or a given alternative) and describe the skewness of a distribution.